

Dynamo Intel XPU K8s Gap Analysis

Dynamo vLLM Backend (Kubernetes DynamoGraphDeployment): Intel XPU Support Gap Analysis

- Already supported on Intel XPU today

- Not supported — no `deploy/xpu/` Kubernetes template exists for any of these

 - Group A — in progress, not a vendor lock-in

 - Group B — no hardware blocker found; simply not yet built (real gap, worth closing)

Summary

Additional gap found on review: `v1beta1` CRD migration is incomplete for Intel XPU

Follow-up

Dynamo vLLM Backend (Kubernetes DynamoGraphDeployment): Intel XPU Support Gap Analysis

Scope: **Kubernetes deployment only** (`examples/backends/vllm/deploy/`). The `bash/CLI launch/` scripts are out of scope for this document per request.

Source: original Confluence-pasted list (`to_fi_original_raw.md`, table formatting lost — see that file for the raw paste) cross-checked directly against the dynamo repo clone at `examples/backends/vllm/deploy/` and `examples/backends/vllm/deploy/xpu/`.

Already supported on Intel XPU today

These 8 have a real, shipped `deploy/xpu/*.yaml` Kubernetes DRA template in the repo — documented in full in `whitepaper/dynamo/01-aggregated.md` through `08-disaggregated-kv-router.md`:

Case	File
Aggregated	<code>deploy/xpu/agg_xpu_dra.yaml</code>
Aggregated + Tracing	<code>deploy/xpu/agg_tracing_xpu_dra.yaml</code>
Aggregated + KV Router	<code>deploy/xpu/agg_router_xpu_dra.yaml</code>
Aggregated + KV Router (no KV events)	<code>deploy/xpu/agg_router_kv_approx_xpu_dra.yaml</code>
Disaggregated	<code>deploy/xpu/disagg_xpu_dra.yaml</code>
Disaggregated + Tracing	<code>deploy/xpu/disagg_tracing_xpu_dra.yaml</code>
Disaggregated + Planner	<code>deploy/xpu/disagg_planner_xpu_dra.yaml</code>
Disaggregated + KV Router	<code>deploy/xpu/disagg_router_xpu_dra.yaml</code>

Plus, outside this specific directory: Global Planner (`examples/global_planner/`) and the heterogeneous Qwen3-VL Intel XPU + NVIDIA GPU recipe (`recipes/qwen3-vl-32b-fp8/`) — see `whitepaper/dynamo/09-global-planner.md` and `10-recipe-qwen3-vl-hetero-xpu-gpu.md`.

Not supported — no `deploy/xpu/` Kubernetes template exists for any of these

None of the 16 cases below have a Kubernetes XPU template today. None are blocked by DRA itself — Intel already has its own DRA driver (`gpu.intel.com`), used by all 8 supported XPU templates. They split into two groups by *why*.

Group A — in progress, not a vendor lock-in

Case	Why not supported
<code>agg_failover</code>	

Case	Why not supported
	Depends on the GMS (GPU Memory Service) sidecar. GMS's own design doc (lib/gpu_memory_service/GMS_MULTI_DEVICE.md) shows CUDA support done and an Intel XPU backend (XpuVMM) planned as “Phase 2,” just not implemented yet. Not a hardware lock-in, just unfinished
agg_gms	Same root cause as agg_failover
gms-failover	Same root cause, inter-pod failover variant

Group B — no hardware blocker found; simply not yet built (real gap, worth closing)

Case	Why not supported
agg_kvbm	Intel XPU support for KVBM is real, active work — not yet merged. Two open GitHub PRs (#7946, #10520, both unmerged as of this writing) add SYCL/Level-Zero KVBM XPU support, tracked by DEP issue #9313 (“kvbm-v2 xpu-sycl enablement”), still in design-discussion stage. Once merged, this K8s template gap should close quickly
disagg_kvbm	Same KVBM caveat as above
disagg_kvbm_2p2d	Same KVBM caveat as above
disagg_kvbm_tp2	Same KVBM caveat as above
disagg-multinode	Uses the same NIXL prefill/decode split already proven working on Intel XPU single-node (deploy/xpu/disagg_xpu_dra.yaml) — just spread across nodes. No hardware-specific field found;

Case	Why not supported
	nobody has built the multi-node XPU variant yet
agg_lora	No K8s XPU template exists, but the equivalent bash launch script already runs on Intel XPU (examples/backends/vllm/launch/lora/xpu/agg_lora_xpu.sh) — strong evidence this is a pure templating gap, not a hardware limitation
agg_qwen_lora	Same reasoning as agg_lora (multimodal LoRA variant) — no XPU-specific dependency identified, just not templated for K8s yet
lora-model	Plain DynamoModel CRD for registering a LoRA adapter — hardware-agnostic manifest, no accelerator-specific field; simply hasn't been paired with an XPU deployment example
minio-secret	Plain Kubernetes Secret for MinIO credentials — hardware-agnostic, works identically regardless of accelerator
sync-lora-job	Plain Kubernetes Job that downloads/uploads adapter weights — hardware-agnostic, no GPU/accelerator step in the job itself
gaie/agg	Gateway API Inference Extension (EPP/InferencePool) integration — the same routing-layer pattern already proven hardware-agnostic on the llm-d side of this whitepaper (Flow Control, Multi-Model Routing); no XPU-specific field found, just not built for this integration point yet
gaie/disagg	Same reasoning as gaie/agg

Case	Why not supported
gaie/http-route	Plain Kubernetes Gateway API HTTPRoute manifest — contains no hardware reference at all; trivially reusable with any backend

Summary

	Count
Already supported (real deploy/xpu/ template exists)	8
Not supported — in progress (GMS/XpuVMM, planned but not yet built)	3 (agg_failover, agg_gms, gms-failover)
Not supported — no blocker found, just not built yet	13

Additional gap found on review: v1beta1 CRD migration is incomplete for Intel XPU

The repo is mid-migration from the `nvidia.com/v1alpha1` CRD schema (`services: map`) to `nvidia.com/v1beta1` (`components: list` — a real structural change, not just a version bump). `deploy/README.md` states plainly: “Equivalent `nvidia.com/v1beta1` templates are available under `v1beta1/`” — implying full parity. **For Intel XPU specifically, this is not true:**

API version	XPU templates present
v1alpha1 (deploy/xpu/)	All 8 (agg, agg_tracing, agg_router, agg_router_kv_approx, disagg, disagg_tracing, disagg_planner, disagg_router)
v1beta1 (deploy/v1beta1/xpu/)	Only 2 (agg_xpu_dra.yaml, disagg_xpu_dra.yaml)

The other 6 Intel XPU templates have **not** been ported to the newer v1beta1 schema yet. This whitepaper's Chapter 1 (`whitepaper/dynamo/01-aggregated.md` through `08-disaggregated-kv-router.md`) documents the v1alpha1 versions (matching what the repo actually ships for all 8 patterns) and did not previously call out this schema-migration gap — worth a short note there so readers on newer Dynamo Platform versions know v1beta1 XPU coverage is currently partial.

Also noted in passing: `lora/` and `lora/multimodal/` each ship their own separate `minio-secret.yaml` / `sync-lora-job.yaml` (not shared) — both still fall under the same “hardware-agnostic, just not paired with an XPU example” reasoning as the single rows already listed above, so this doesn't change the Group B classification, just the file count.

Follow-up

The 13 “Group B” cases are the actionable backlog: building and validating any of them on real Intel XPU hardware, then contributing a `deploy/xpu/` template upstream, would close a real gap. The LoRA cases (`agg_lora`, `agg_qwen_lora`) are the lowest-risk starting point since the underlying capability is already proven working on Intel XPU via the `bash` launch-script path.